Temperature, Heat and 1st Law of Thermodynamics

Temperature 0th Law of Thermodynamics
Thermal expansion Heat and Work
1st Law of Thermodynamics

Chapter 19:Temperature

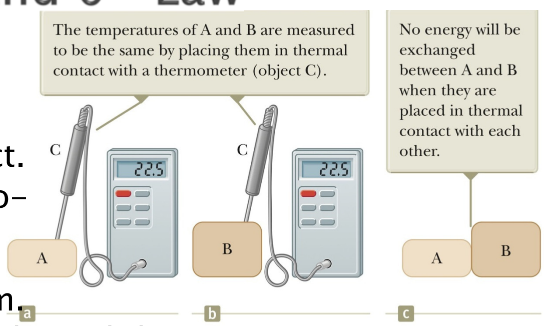
Temperature and 0th Law of Thermodynamics
Thermometers and Celsius Scale
Constant volume gas thermometer
Thermal expansion of solids and liquids
Macroscopic description of an ideal gas

Temperature

- Why does a tile floor feel colder than a carpet on a tile floor – even though they are at the same temperature.
- Our skin measures the rate of transfer of heat – Tile transfers heat faster than the carpet.
- Need a more reliable, reproducible method to measure temperature.
- Thermal equilibrium – two objects (can be at different temperatures) in thermal contact with each other will eventually reach the same final temperature due to energy exchanged between the two objects.

Temperature and 0th Law

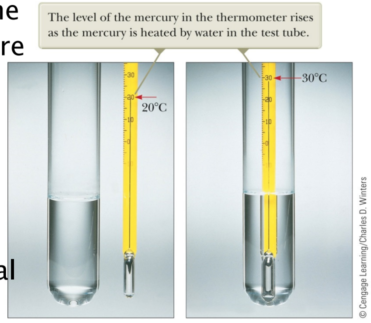
- Consider two objects A and B.
- Not in thermal contact.
- Third object C thermo–meter, is in contact with each object until in thermal equilibrium.

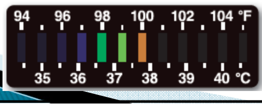


- If the thermometer indicated the same temperature for both, then objects A and B are in thermal equilibrium.
- Zeroth Law: If objects A and B are separately in thermal equilibrium with a third object C, then objects A and B are in thermal equilibrium with each other.

Temperature and Celsius Scale

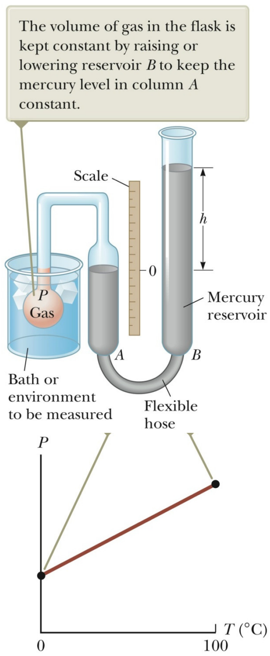
- Some physical properties that change with temperature:
  - Volume of liquid
  - Dimensions of a solid
  - Pressure of a gas at constant volume
  - Volume of a gas at constant pressure
  - Electrical resistance of a conductor
  - Colour of and object.
- Common thermometers
  - Calibrate on Celsius scale
  - 0°C water and ice in thermal equil.
  - 100°C – water and steam in thermal equil.





Constant-Volume Gas Thermometer

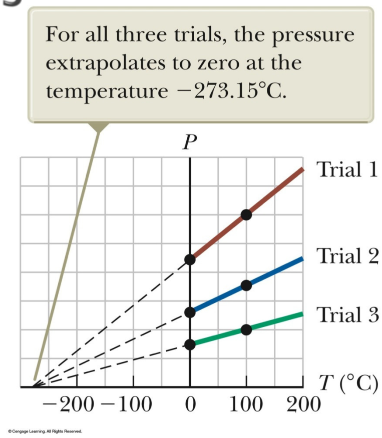
- Flask immersed in ice-water
- Adjust column A to the zero height.
- Measure the height h difference between column A and B
- Indication of the pressure in flask.
- $P = P_0 + \rho g h$
- Immerse flask in boiling water.
- Adjust column A to the zero height, and measure the height h difference between column A and B
- Draw staight line graph betwen 2 points and read off any temperature.



Constant-Volume Gas Thermometer

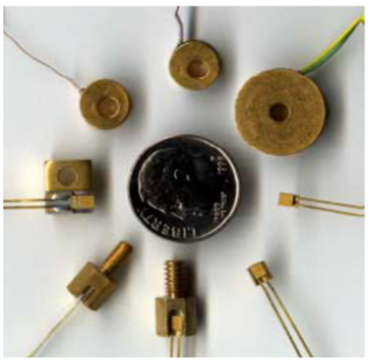
- Different gases
  - Between 0°C and 100°C
  - Extrapolate below 0°C
  - Pressure is zero when temperture is **−273.15°C.**
  - **Absolute zero** on the absolute temperature scale.
- SI unit: Kelvin – K
- Other units: $T_C = (T_K - 273.15)$

$$T_F = \frac{9}{5}T_C + 32°$$



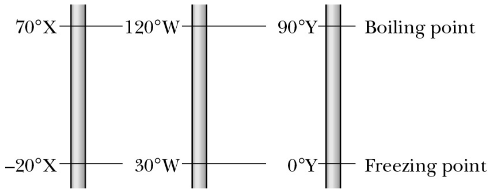
Thermal properties

- Thermocouples



Problem

- The figure shows 3 linear temperature scales with freezing and boiling points of water indicated. (a) Rank the degrees on these scales by size. (b) Rank the following temperatures highest first: 50 °X, 50°W and 50°Y.

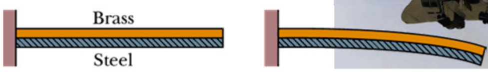


Thermal Expansion

- What happens when the temperature of a material increases?
  - Energy is added
  - Atoms move a bit further apart
  - Different materials' atoms move different distances further apart

Thermal Expansion

- Thermal expansion must be taken into account in everyday situations or used to our advantage.
  - Bridge, train tracks and buildings have expansion joints
  - Dental fillings
  - Concorde (due to frictional forces)
  - Bimetal strip
  - Thermostats
  - Thermometers



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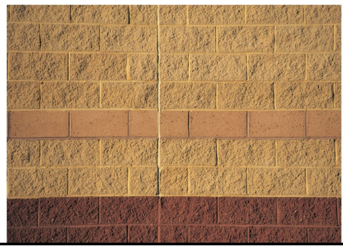

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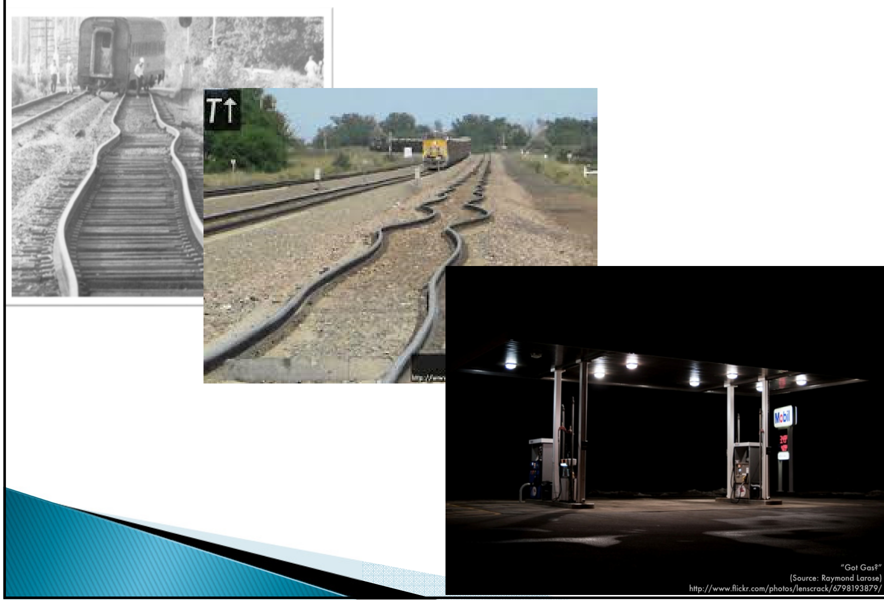
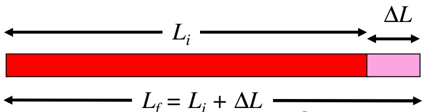
Thermal Expansion

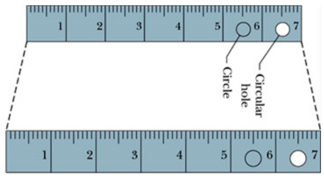


Linear Expansion

- Consider a metal rod of length L which is raised by a temperature ΔT.



- Its length will increase from L_i to L_f : $\Delta L = L_i \alpha \Delta T$
- α is the coefficient of linear expansion
- Units of α : °C^{-1} or K^{-1}
- How do holes and voids in objects expand?
  - Holes
  - Circle
  - Scale
  - Numbers



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Volume Expansion

- All dimensions increase with an increase in temperature.
- Volume must also increase.
- $\Delta V = V \beta \Delta T$
- β is the coefficient of volume expansion
- $\beta = 3\alpha$
- Table 19.1

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Thermal expansion

Table 19.1 Average Expansion Coefficients for Some Materials Near Room Temperature

Material (Solids)	Average Linear Expansion Coefficient $(\alpha)(^\circ C)^{-1}$	Material (Liquids and Gases)	Average Volume Expansion Coefficient $(\beta)(^\circ C)^{-1}$
Aluminum	24×10^{-6}	Acetone	1.5×10^{-4}
Brass and bronze	19×10^{-6}	Alcohol, ethyl	1.12×10^{-4}
Concrete	12×10^{-6}	Benzene	1.24×10^{-4}
Copper	17×10^{-6}	Gasoline	9.6×10^{-4}
Glass (ordinary)	9×10^{-6}	Glycerin	4.85×10^{-4}
Glass (Pyrex)	3.2×10^{-6}	Mercury	1.82×10^{-4}
Invar (Ni–Fe alloy)	0.9×10^{-6}	Turpentine	9.0×10^{-4}
Lead	29×10^{-6}	Air[a] at 0°C	3.67×10^{-3}
Steel	11×10^{-6}	Helium[a]	3.665×10^{-3}

[a]Gases do not have a specific value for the volume expansion coefficient because the amount of expansion depends on the type of process through which the gas is taken. The values given here assume the gas undergoes an expansion at constant pressure.

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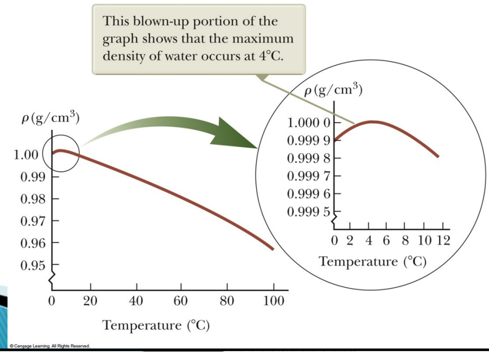
Why does ice float in water?

- Above 4 °C water expands with an increase in temperature like other solids – density decreases.
- Between 0 to 4 °C water contracts with increase in temperature.
- At 4 °C water has a max. density.



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Problem

- A steel train track has a length of 30 m when the temperature is 32 °F. What is its length when the temperature is (a) 313 K? (b) –40°C?

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Problem

- On a hot day in LA, an oil trucker loaded 37 000 L of diesel fuel. He encountered cold weather on his journey, and where it was 23.0 K lower than in LA, he delivers his entire load. How many litres did he deliver? The coefficient of volume expansion for diesel is 9.50×10^{-4}/C°, and the coefficient of linear expansion for the steel tank is 11×10^{-6}/C°.

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Macroscopic description of an ideal gas

- Read section 19.5

Chapter 20: 1st Law of Thermodynamics

Heat and internal energy
Specific heat
Latent heat
Work and heat in thermodynamic processes
1st Law of thermodynamics
Energy transfer mechanisms

Heat and internal energy

- Internal energy is all the energy of a system that is associated with its microscopic components – atoms and molecules – when viewed from a reference frame at rest with respect to the com of the system.

- Heat Q: A Process of transferring energy across the boundary of a system due to a temperature difference between the system and its surroundings.
- Heat is also the amount of energy, Q, transferred by this process – measured in Joule
- Other common units:
  - 1 calorie (cal) = 4.1860 J
  - 1 Calorie (Cal) = 1kcal = 4 186.0 J [Diet]

J.P. Joule 1818 – 1889

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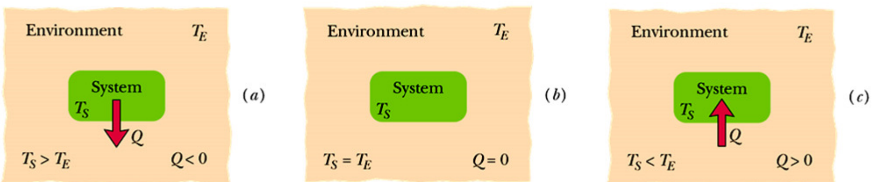
Heat

- $T_S > T_E$
- $Q < 0$

$T_S = T_E$
$Q = 0$

$T_S < T_E$
$Q > 0$



Sign convention:
Flow out of system: $Q < 0$
Flow into system: $Q > 0$

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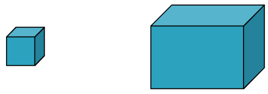
Specific Heat and Calorimetry

- Heat capacity C – proportionality constant (J/K) between Q and ΔT

$$Q = C\Delta T = C\left(T_f - T_i\right)$$



- Specific Heat (Capacity) c – Heat capacity per unit mass (J/kg·K)

$$Q = cm\Delta T = cm\left(T_f - T_i\right)$$

Table 20.1

Specific Heat and Calorimetry

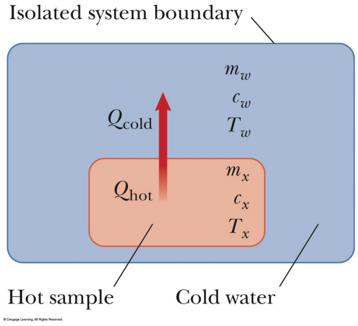
Table 20.1 Specific Heats of Some Substances at 25°C and Atmospheric Pressure

Substance	Specific Heat (J/kg · °C)	Substance	Specific Heat (J/kg · °C)
Elemental solids		*Other solids*	
Aluminum	900	Brass	380
Beryllium	1 830	Glass	837
Cadmium	230	Ice (−5°C)	2 090
Copper	387	Marble	860
Germanium	322	Wood	1 700
Gold	129		
Iron	448	*Liquids*	
Lead	128	Alcohol (ethyl)	2 400
Silicon	703	Mercury	140
Silver	234	Water (15°C)	4 186
		Gas	
		Steam (100°C)	2 010

Note: To convert values to units of cal/g · °C, divide by 4 186.

Specific Heat and Calorimetry

- Calorimetry –
  - Heat an object to a specific temperature T_x
  - Place the object in a container of with a known amount of water at a known temperature $T_x > T_w$
  - Measure the final equilibrium temperature of the water.
- $Q_{cold} = -Q_{hot}$
- $m_w c_w \left(T_f - T_w\right) = -m_x c_x \left(T_f - T_x\right)$
- Solve for the unknown c_x



Example

- A certain amount of heat Q will warm 1g of material A by 3 C° and 1 g of material B by 4 C°. Which material has the greater specific heat?

Example 20.3

A cowboy fires a silver bullet with a muzzle speed of 200 m/s into a wooden wall. Assume all the internal energy generated by the impact remains with the bullet. What is the temperature change of the bullet?

Latent Heat

- Latent Heat, L – Heat absorbed by the sample when it changes phase – temperature is constant (J kg^{-1})

$$Q = Lm$$

  - L_F heat of fusion (freezing or melting)
  - L_V heat of vaporisation (boiling or condensing)

Table 20.2

Latent Heat

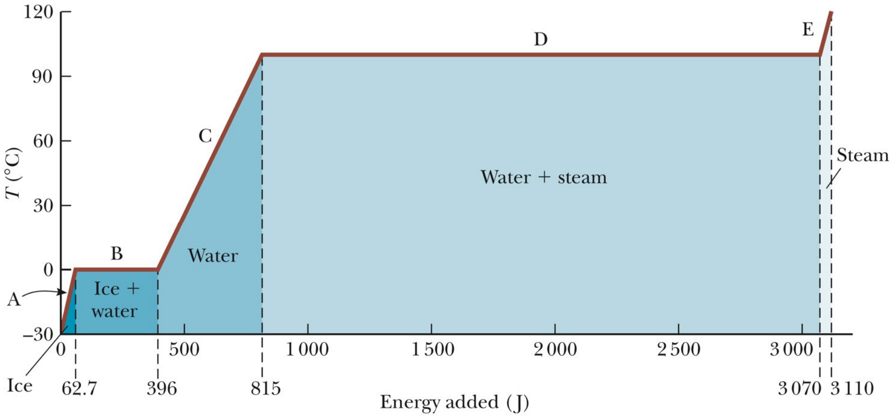
Table 20.2 Latent Heats of Fusion and Vaporization

Substance	Melting Point (°C)	Latent Heat of Fusion (J/kg)	Boiling Point (°C)	Latent Heat of Vaporization (J/kg)
Helium[a]	−272.2	5.23×10^3	−268.93	2.09×10^4
Oxygen	−218.79	1.38×10^4	−182.97	2.13×10^5
Nitrogen	−209.97	2.55×10^4	−195.81	2.01×10^5
Ethyl alcohol	−114	1.04×10^5	78	8.54×10^5
Water	0.00	3.33×10^5	100.00	2.26×10^6
Sulfur	119	3.81×10^4	444.60	3.26×10^5
Lead	327.3	2.45×10^4	1 750	8.70×10^5
Aluminum	660	3.97×10^5	2 450	1.14×10^7
Silver	960.80	8.82×10^4	2 193	2.33×10^6
Gold	1 063.00	6.44×10^4	2 660	1.58×10^6
Copper	1 083	1.34×10^5	1 187	5.06×10^6

[a]Helium does not solidify at atmospheric pressure. The melting point given here corresponds to a pressure of 2.5 MPa.

Latent Heat

- 1 g ice from −30ºC to 120ºC.



Problem

- How much heat must be absorbed by ice of mass $m = 720$ g at -10 °C to take it to liquid state at 15°C?

Problem

- If we supply the ice with a total of only 210 kJ (as heat), what are the final temperature and state of the water?:

Problem

- A copper slug whose mass m_c is 75 g is heated in a lab oven to 312°C. The slug is then dropped into a glass beaker containing a mass m_w = 220 g of water. The heat capacity C_b of the beaker is 45 cal/K .The initial temperature T_i of the water and the beaker is 12 °C. Assuming the slug, beaker and the water are an isolated system and the water does not vaporize, find the final temperature T_f of the system at thermal equilibrium.

45 cal/K = 188.41 J/K

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Problem

	Copper	water	beaker
m (g)	74g	220 g	
C_b (J/K)			188.41
C (J/kg.K)	389	4180	
T_i (°C)	312	12	12
T_f (°C)	?	?	?

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Heat and work in a thermodynamic process

- State of a system is described by State Variables
  - Pressure
  - Volume
  - Temperature
  - Internal Energy
- State of a system can only be described when it is in thermal equilibrium.
- Transfer-Variables

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{MW} + T_{MT} + T_{ET} + T_{ER}$$

  - RHS: these variables cause energy to be transferred across the system's boundaries
  - Energy transfer: + in and – out of the system.

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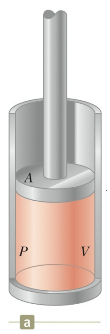
Heat and work in a thermodynamic process

- Ch 7 – Word done on a particle or object
- Now consider the work done on a deformable system – e.g. a gas
  - At equilibrium – gas has a volume V
  - The gas exerts a pressure P on the sides of the container and the piston
  - Piston has an areas, A and therefore $F = PA$
  - Upward force on the piston due to the gas is then = weight of the piston on the gas.



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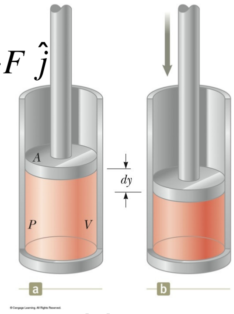
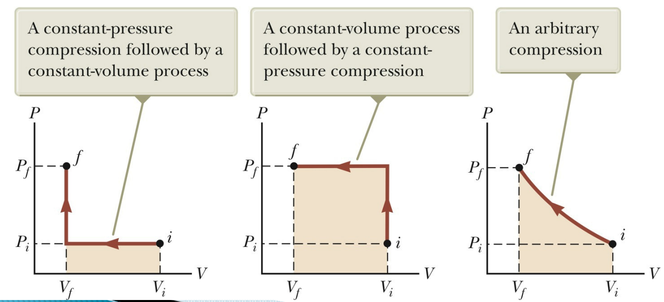
Heat and work in a thermodynamic process

- If the piston is pushed down slowly – quasi-static thermal equilibrium the whole time.
- The force on the gas downwards $\vec{F} = -F\,\hat{j}$
- Displacement $d\vec{r} = dy\,\hat{j}$
- Work done on the gas:
  $dW = \vec{F} \cdot d\vec{r} = -F\,\hat{j} \cdot dy\,\hat{j} = -PAdy$
- $Ady = dV$
- $dW = -PdV$
- Gas is compressed – dV is negative, W is positive.
  Gas expands – dV is positive, W is negative.
- $W = -\int PdV$, and P is usually dependent on V and T



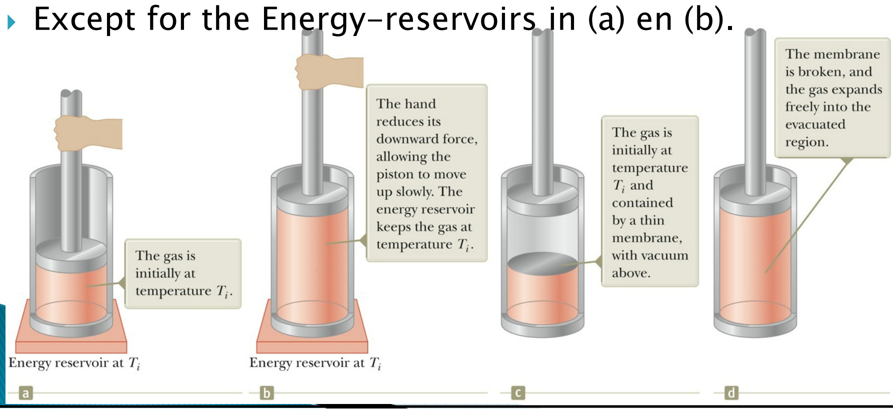
Heat and work in a thermodynamic process

- PV diagrams
- Lots of ways for a system to change from an initial state i to a final state f – This is given by P-V curve
- Work done on the gas going from state i to state f is given by the negative area under the curve
- Work is path –dependent



Heat and work in a thermodynamic process

- Energy transferred in or out of a system in the form of heat, Q, depends on the process used.
- Ideal gas at V_i, P_i, T_i – thermally insulated on the sides of the containers.
- Except for the Energy-reservoirs in (a) en (b).



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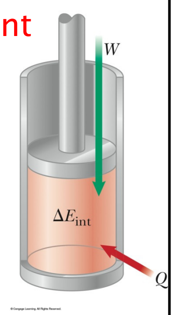
1st Law of Thermodynamics

- Ch 8 – Conservation of energy.
- When the state of a system changes from i to f, Q and W are dependent on the type of process used.
- BUT $Q + W$ is always the same for the different processes – path independent.
- $Q + W = \Delta E_{\text{int}} = E_{\text{int},f} - E_{\text{int},i}$
- First law of thermodynamics



- An isolated system (no interaction with the environment)
  - No transfer of energy in the form of heat
  - No work done on the system

$$Q + W = \Delta E_{\text{int}} = 0$$
$$\therefore E_{\text{int},f} = E_{\text{int},i}$$

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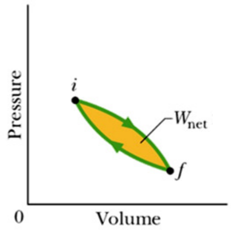
1^{st} Law of Thermodynamics

- Cyclic process (process which starts and ends in the same state)
  - Change in internal energy is zero.
  - Therefore

$$Q + W = \Delta E_{\text{int}} = 0$$

$$Q = -W$$



Applications of the 1^{st} Law

- Adiabatic Process: $Q = 0$, $\Delta E_{\text{int}} = W$ eg. If the gas is compressed work is done (W is +ve) and the change in the internal energy is positive.
- Free expansion: $W = 0$, $Q = 0$, $\Delta E_{\text{int}} = 0$ i.e. adiabatic process where no heat is transferred, and no work is done on or by the system.
- Iso-volumetric process: $W = 0$, $\Delta E_{\text{int}} = Q$ i.e. $\Delta V = 0$, heat is absorbed or lost by the system.
- Iso-baric process: $W = -P(V_f - V_i)$, $\Delta E_{\text{int}} = Q + W$ eg. The system has a constant pressure through-out the process (Fig 20.6 a and b).

Applications of the 1st Law

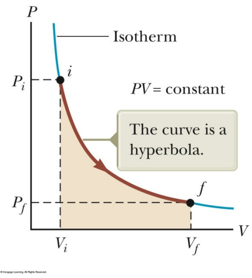
- Iso-thermal process: $\Delta T = 0,\ \Delta E_{\text{int}} = 0$

  $\therefore Q = -W$

- $$W = -\int_{V_i}^{V_f} P\,dV = -\int_{V_i}^{V_f} \frac{nRT}{V}\,dV$$

  $$W = -nRT\int_{V_i}^{V_f} \frac{dV}{V} = -nRT \ln V\Big|_{V_i}^{V_f}$$

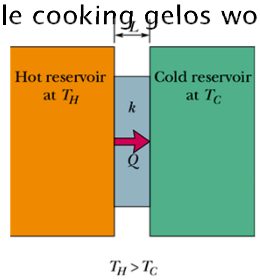
  $$= -nRT \ln \frac{V_i}{V_f}$$



  - eg. Any energy that enters the system by heat is transferred out of the system by work; as a result, no change in the internal energy of the system occurs (external force does negative work).

Energy transfer mechanisms

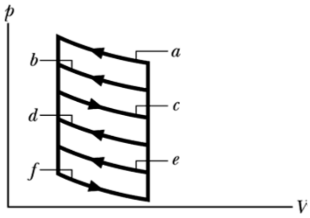
- Thermal conduction
  - Metal spoon on a let in the pot while cooking gelos word



- Convection
  - Heater in a room
- Radiation
  - Energy from the sun

Example

- The p–V diagram here shows 6 curved paths (connected by vertical paths) that can be followed by a gas. Which 2 of the curved paths should be part of a closed cycle (those curved paths plus connecting vertical paths) if the net work done on the gas during the cycle is to be at its
- (a) maximum negative value?
- (b) maximum positive value?



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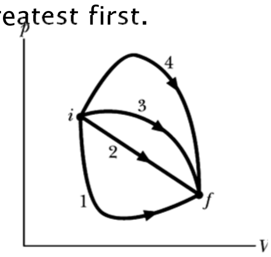
Example

- The figure shows 4 paths on the p-V diagram along which a gas can be taken from state i to f. Rank the paths according to (a) the change ΔE_{int} of the gas, (b) the work done by the gas, and (c) the magnitude of the energy transferred as heat Q between gas and the environment, greatest first.



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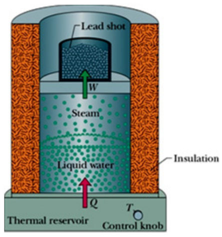
Problem



Let 1.00 kg of liquid water be at 100°C be converted to steam at 100°C by boiling at standard atmospheric pressure (1.00 atm = 1.01×10^5 Pa) in an arrangement shown. The volume of the water changes from 1.00×10^{-3} m^3 as a liquid to 1.671 m^3 as steam.

(a) How much work is done on the system during this process?

(b) How much energy is transferred as heat during the process?

(c) What is the change in the system's internal energy?

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Example 20.5 and 20.6

- Similar to above.

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Example 20.7

1.0-kg bar of copper is heated at atmospheric pressure so that its temperature increases from 20°C to 50°C.

- What is the work done on the copper bar by the surrounding atmosphere?
- How much energy is transferred to the copper by heat?
- What is the increase in the internal energy of the copper bar?